

← Go Back

Hardware

UNO R4 WiFi

Tutorials

Arduino UNO R4 WiFi ADC Resolution

Arduino UNO R4 WiFi CAN Bus

Arduino UNO R4 WiFi User Manual

Arduino UNO R4 WiFi Arduino Cloud Setup Guide

Arduino UNO R4 WiFi Digital-to-Analog Converter (DAC)

Debugging the Arduino UNO R4 WiFi

Arduino UNO R4 WiFi EEPROM

UNO R4 WiFi Custom Firmware Upload to ESP32 (Advanced)

Using the Arduino UNO R4 WiFi LED Matrix

Arduino UNO R4 WiFi OPAMP

Arduino UNO R4 WiFi Qwiic Connector

Getting Started with UNO R4 WiFi

Arduino UNO R4 WiFi Real-Time Clock

UNO R4 WiFi Stack Trace Debug Runtime Errors

UNO R4 Capacitive-Touch Tutorial

Arduino UNO R4 WiFi USB HID

Home / Hardware / UNO R4 WiFi / Arduino UNO R4 WiFi EEPROM

Arduino UNO R4 WiFi EEPROM

Learn how to access the EEPROM memory on the UNO R4 WiFi.

Author · Karl Söderby · Last revision · 10/07/2025

In this tutorial you will learn how to access the EEPROM (memory) on an **Arduino UNO R4 WiFi** board. The EEPROM is embedded in the UNO R4 WiFi's microcontroller (RA4M1).

Goals

The goals of this tutorials are:

- Write to the EEPROM memory,
- read from the EEPROM memory.

Hardware & Software Needed

- Arduino IDE ([online](#) or [offline](#))
- [Arduino UNO R4 WiFi](#)
- [UNO R4 Board Package](#)

EEPROM

Electronically erasable

ON THIS PAGE

Goals

- Hardware & Software Needed
- EEPROM +
- Summary

be used to store data that can be retrieved after power loss - it is non-volatile. EEPROM memory can be useful during run-time to log data, or can be used to re-initialize variables whenever a system comes back online.

When writing to the EEPROM memory, we specify two parameters: the **address** and **value**. Each byte can hold a value between 0-255.

```
1 EEPROM.write(0,15); //w
```

We are writing a value of `15` to the first byte available in the memory, `0`.

To read the value of this memory, we simply use:

```
1 EEPROM.read(0); //reads
```

There are several more methods available when working with EEPROM, and you can read more about this in the [A Guide to EEPROM](#).



Please note: EEPROM is a type of memory with a limited amount of write cycles. Be cautious when writing to this memory as you may significantly reduce the lifespan of this memory.

A minimal example on how to **write** to the EEPROM can be found below:

```
1  #include <EEPROM.h>
2
3  int addr = 0;
4  byte value = 100;
5
6  void setup() {
7      EEPROM.write(addr, value);
8  }
9  void loop(){
10 }
```

EEPROM Read

A minimal example of how to **read** from the EEPROM can be found below:

```
1  #include <EEPROM.h>
2
3  int addr = 0;
4  byte value;
5
6  void setup() {
7      Serial.begin(9600);
8      value = EEPROM.read(addr);
9      while (!Serial) {
10
11      }
12
13      Serial.print("Address: ");
14      Serial.println(value);
15  }
16
17  void loop() {
18  }
```

Summary

In this tutorial, you've learned how to access the EEPROM on the UNO R4 WiFi board. To learn more about EEPROM, visit [A Guide to EEPROM](#).

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